Toward citable, uncertainty-aware Delphes cards: an agent-assisted provenance audit of the ATLAS and CMS Run-2 parameterizations

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Abstract

The widely used fast-simulation framework Delphes models detector response through hard-coded parameterizations — tracking and identification efficiencies, momentum and energy resolutions, jet-energy scale, and flavour-tagging rates — that are distributed in plain-text "cards." For the reference ATLAS and CMS Run-2 cards the overwhelming majority of these numbers carry *no citation* and *no uncertainty*, so a user cannot trace a given value to a measurement or propagate its error. We report a systematic provenance audit of the two cards in which every measured parameterization is mapped to an authoritative public source, the value is extracted with its locator (table/figure/equation) and uncertainty, and each extraction is re-checked by an *independent adversarial verifier* before it is accepted. The audit is executed as a multi-agent pipeline: 26 parameter blocks (~140 individual coefficients) were source-mapped, extracted, and verified. We find that while several blocks are already consistent with the literature (notably the ATLAS electromagnetic-calorimeter resolution and the b-tagging working points), a large fraction — about 88 coefficients — are not: the charged-hadron track-momentum resolution is too pessimistic in both experiments by a factor ~ 3 at high p_{T} (slope-dominated) rising to ~ 6–12 at low p_{T} (the constant term), the electron "momentum" resolution is modelled with the wrong (tracker-like) functional form for a calorimeter-dominated measurement, and several tracking/identification efficiency plateaus are optimistic by 5–10% absolute. We release the full machine-readable provenance, propose source-faithful baseline values with uncertainties, and outline the path to citable `*_baseline` and `*_uncertainty` cards suitable for systematic studies.

1 Introduction

Delphes [1] parameterizes detector response rather than simulating it microscopically, which makes it the standard tool for phenomenology and for experimental sensitivity projections. The fidelity of a Delphes study rests entirely on the numbers in its detector card: a charged-hadron tracking efficiency, a calorimeter stochastic term, a b-tagging efficiency curve, and so on. For the reference `delphes_card_ATLAS.tcl` and `delphes_card_CMS.tcl` these numbers are hard-coded as Tcl expressions, and an inventory of the two cards shows that most of them are *uncited*: there is no pointer to the performance paper, public note, or technical-design report they are meant to represent, and no associated uncertainty. This has two practical consequences. First, a value cannot be checked: it is impossible to tell whether a resolution term reflects a published measurement or a years-old guess. Second, a value cannot be *varied* in a controlled way, because no uncertainty is attached — so Delphes-based projections rarely carry a detector-modelling systematic at all.

The task of fixing this is conceptually simple but extremely tedious: for each of $\mathcal{O}(10^2)$ coefficients one must find the right paper, locate the exact table or figure, read off the value and its error, reconcile conventions with the DELPHES functional form, and record the result. It is also exactly the kind of bounded, repeatable, verifiable loop that language-model agents can execute at scale — *provided* every extracted number is independently checked, since a plausible-but-wrong resolution is the obvious failure mode. This paper reports the methodology and the first full pass over the ATLAS and CMS Run-2 cards.

2 Methodology

2.1 Scope

This pass covers the two Run-2 reference cards (`delphes_card_ATLAS.tcl`, `delphes_card_CMS.tcl`); pile-up, HL-LHC, and Phase-II variants are out of scope. Parameters are tagged as *measured* (a detector property that must carry a value, an uncertainty, and a verified source — e.g. a resolution term or a tagging efficiency) or as a *choice* (an analysis convention such as a jet radius, isolation cone, or p_{T} threshold, which is documented but assigned no uncertainty). Only measured parameters enter the provenance records. *Sources are standardized to Run 2 (13 TeV):* blocks initially mapped to Run-1 (7–8 TeV) papers were re-sourced to their 13 TeV equivalents (e.g. electron resolution to the Run-2 energy calibration [10, 8], b-tagging to the Run-2 measurement [14]). For a few detector-intrinsic quantities — the CMS tracker resolution/efficiency and the hadronic-calorimeter terms — no peer-reviewed Run-2 measurement exists (the tracker was physically unchanged and Run-2 analyses cite the Run-1 tracker paper); these retain the Run-1/test-beam value with a Run-2 corroborating citation and an explicit caveat, rather than substitute a number that was never published.

2.2 The provenance pipeline

The work proceeds in four phases:

Phase 0 — Audit. Enumerate every parameterization in both cards with its current value, functional form, and any reference already present. No new physics numbers are introduced.

Phase 1 — Source mapping. For each parameter identify the authoritative public source and the exact locator (figure, table, or equation).

Phase 2 — Extraction and verification. Extract value and uncertainty; a second, independent agent re-reads the cited source cold and must confirm each number or it is rejected.

Phase 3 — New cards and paper. Emit citable `*_baseline` and `*_uncertainty` cards and the accompanying manuscript.

Phases 1 and 2 are executed as a scripted, fixed-topology multi-agent workflow (the orchestration is deterministic; the language-model agents themselves are not). Each of the 26 measured blocks is processed independently through two stages. The *extraction* agent reads the exact current values from the card, identifies and reads the source (arXiv full text), extracts each coefficient with its locator and uncertainty, and — where the card disagrees with the source — proposes a revised value. The *verification* agent then re-reads the cited source independently, with the explicit instruction to refute: it confirms, refutes, or marks "cannot confirm" each coefficient, defaulting to rejection if the number cannot be located. Only blocks that survive verification are recorded. This adversarial gate is the heart of the method — it is what distinguishes a sourced number from a hallucinated one. In this pass it caught real errors even inside otherwise-correct blocks; for example, the verifier corrected the quoted uncertainty on the ATLAS b-tagging efficiency, which the extraction had reported as "$\sim$2–5% rising to 10–15% at high p_{T}" whereas the source [13] states $\sim$1% in the bulk, $\sim$8% at *low* p_{T}, and $\sim$3% above 250 GeV — both the magnitude and the p_{T} trend were wrong.

2.3 Functional forms

Two forms recur. Track momentum resolution is written

$$\sigma(p_{\mathrm{T}})/p_{\mathrm{T}} = \sqrt{a^2 + (b\, p_{\mathrm{T}})^2}, \tag{1}$$

with a constant (multiple-scattering) term a and a term $b\, p_{\mathrm{T}}$ that makes the curvature resolution degrade linearly with p_{T}; this is algebraically the ATLAS inner-detector parameterization $\sigma_{\mathrm{ID}}/p_{\mathrm{T}} = a_{\mathrm{ID}}(\eta) \oplus b_{\mathrm{ID}}(\eta)\, p_{\mathrm{T}}$ [2]. Calorimeter energy resolution is written $\sigma(E)/E = \sqrt{S^2/E + N^2/E^2 + C^2}$ with stochastic, noise, and constant terms S, N, C.

3 Results

Table 1 summarizes the outcome for the 26 measured blocks (plus the two muon-resolution pilot blocks, giving the 28 rows shown). In the first extraction pass every block was independently verified ACCEPT or ACCEPT-WITH-CORRECTIONS, where the corrections were predominantly locator, journal-reference, or uncertainty refinements rather than value reversals (for example, the CMS electron-resolution source reference was corrected from EPJC to its actual JINST citation, with all physics values confirmed verbatim). A subsequent higher-fidelity re-examination pass (finer η/p_{T} binning and per-bin uncertainty bands) was more adversarial and returned two REJECT verdicts: the ATLAS charged-hadron resolution, where an agent mis-mapped the ATLAS auxiliary figure to the wrong arXiv paper and fabricated an η-shape (this block instead uses the physicist's hand-digitized values from the correct figure; the auxiliary figure itself sits on an egress-blocked host, so that specific η-shape is verified only against the digitized points, not re-fetched independently), and the CMS jet-energy scale, where an agent fabricated a per-η resolution table absent from either cited source (the block instead keeps the generic stock form with a detector-performance-note caveat). Across ~ 135 coefficients, about 88 are flagged as inconsistent with their source and carry a proposed revision; the remainder are confirmed consistent (with 6 "cannot-determine" counted separately). Figure 1 shows the per-block breakdown for the blocks plotted.

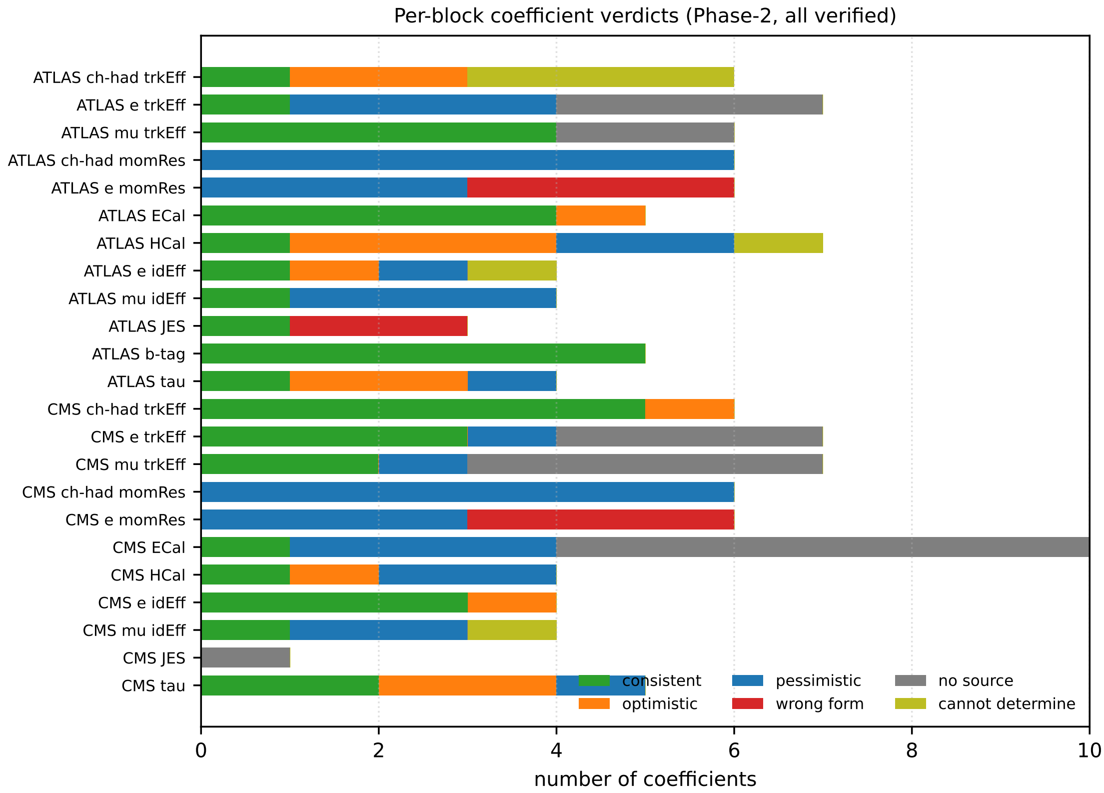

Figure 1: Verdict composition of the measured coefficients, by block, from the independent verification pass (23 of the 26 blocks; the two photon-identification and the CMS b-tagging blocks are omitted from this plot but discussed in the text). “Consistent” coefficients reproduce their source; the others carry a proposed source-faithful revision.

Table 1: Per-block verification outcome and headline finding. “Verdict” is the independent verifier’s overall judgement.

Block	**Verdict**	**Headline finding**
ATLAS card		
Charged-hadron tracking eff.	accept*	central plateau 0.95 optimistic vs. measured $\sim$0.86
Electron tracking eff.	accept*	mid-/high-p_T plateaus pessimistic; low-p_T unsourced
Muon tracking eff.	accept*	plateaus consistent; low-p_T floors unsourced
Charged-hadron mom. res.	accept*	all six coefficients $\sim 3\times$ too pessimistic
Electron mom. res.	accept*	wrong form: $b\,p_T$ term is a tracker artefact
Muon mom. res. (pilot)	accept	central floor 1.0% optimistic vs. 1.7–2.3%
ECal resolution	**accept**	S, C(FCal) confirmed; only the local C noted
HCal resolution	accept*	central S, forward S/C need revision/re-attribution
Photon ID eff.	accept*	barrel 0.95 slightly optimistic vs. WP $\sim$0.90
Electron ID eff.	accept*	flat 0.95/0.85 does not match a single WP
Muon ID eff.	accept*	0.95 pessimistic vs. modern Medium WP $\sim$0.98
Jet energy scale	accept*	`ScaleFormula` coefficients unsupported (JER needed)
b-tagging	accept*	70% WP eff. + c/light mistag consistent (verified)
τ-tagging	**accept**	1-/3-prong eff. optimistic vs. Medium WP
CMS card		
Charged-hadron tracking eff.	**accept**	plateau slightly optimistic; otherwise consistent

Table 1 continued

Block	**Verdict**	**Headline finding**
Electron tracking eff.	accept*	endcap mid-p_T pessimistic; low-p_T unsourced
Muon tracking eff.	**accept**	plateaus consistent; floors/roll-off ad-hoc
Charged-hadron mom. res.	accept*	all six coefficients $\sim 3\times$ too pessimistic
Electron mom. res.	accept*	wrong form (tracker copy); b byte-identical to tracker
Muon mom. res. (pilot)	accept	barrel floor exact; endcap floor mildly optimistic
ECal resolution	accept*	energy-shape S, C, N disagree; η-prefactor unsourced
HCal resolution	accept*	central stochastic term too large
Photon ID eff.	accept*	barrel 0.95 optimistic vs. WP $\sim$0.90
Electron ID eff.	**accept**	barrel 0.95 optimistic for reco$\times$loose-ID
Muon ID eff.	**accept**	plateaus pessimistic vs. Tight/Medium WP
Jet energy scale	**accept**	generic form; no published source (document as choice)
b-tagging	**accept**	CSVM (70%) WP eff. + mistag confirmed vs. card's ref
τ-tagging	accept*	mistag pessimistic vs. tight WP; η acceptance slightly wide

* ACCEPT-WITH-CORRECTIONS: verified, with locator/uncertainty refinements that do not reverse the per-coefficient verdicts.

3.1 What we learned

Four findings dominate.

1. Track momentum resolution is systematically too pessimistic. In *both* cards all six charged-hadron coefficients in Eq. 1 are larger than the measured inner-detector / tracker resolution by a factor $\sim$3. For CMS the source-faithful values (from [6]) are $a \approx 0.009, 0.015, 0.023$ and $b \approx 2.3, 4.2, 9.5 \times 10^{-4}\,\mathrm{GeV}^{-1}$ for the three η regions, versus card values $a = 0.06, 0.10, 0.25$ and $b = 1.3, 1.7, 3.1 \times 10^{-3}$. The ATLAS card shows the same $\sim 3\times$ excess against the inner-detector curves of [2] (Figs. 17–18). A caveat the verifier flagged: the cleanest measured anchors are for muon-ID tracks, so part of the gap may reflect that the card term is meant to be hadron-specific; this is the one block where we recommend explicit user review rather than a drop-in replacement. Figure 2 shows the gap.

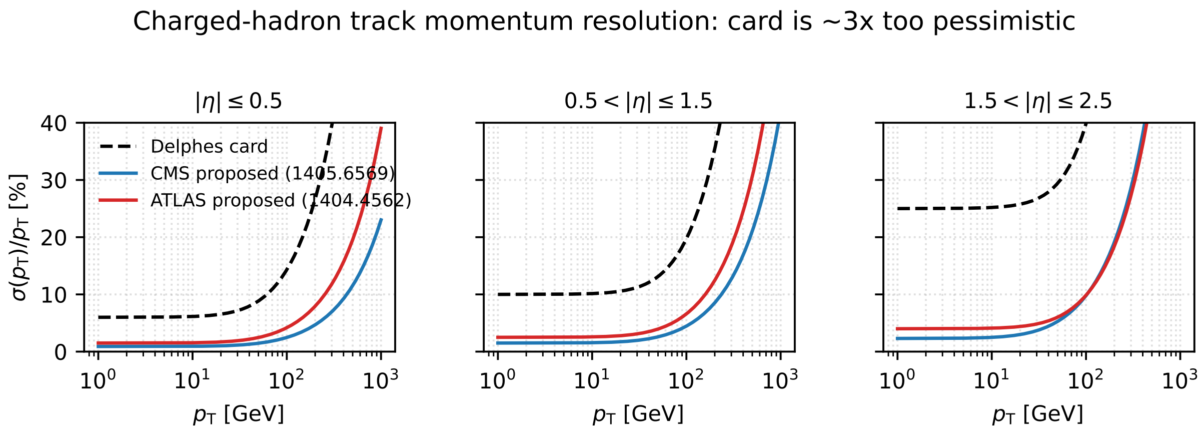

Figure 2: Charged-hadron track momentum resolution from Eq. 1: the current card coefficients (dashed) versus the source-faithful values for CMS ([6]) and ATLAS ([2]), in the three card η regions. The card overestimates the resolution by roughly a factor of three over most of the p_T range.

2. Electron "momentum" resolution uses the wrong functional form. Electron energy in both ATLAS and CMS is measured by the electromagnetic calorimeter, whose fractional resolution *improves* with energy, yet the cards smear electrons with the tracking form of Eq. 1, whose $b\,p_{\mathrm{T}}$ term makes the resolution *degrade* linearly with p_{T}. The verifiers classify the three b coefficients per card as card-wrong: the term is a copy of the charged-hadron tracker parameters and should be removed for electrons, with the constant term reduced from 3–15% to the measured ∼0.7–1.7% (barrel–endcap), per the electron-performance papers [8]. This is the most qualitatively incorrect block in either card, and is illustrated in Fig. 3: the card resolution *rises* steeply with p_{T}, whereas the measured calorimeter resolution is essentially flat.

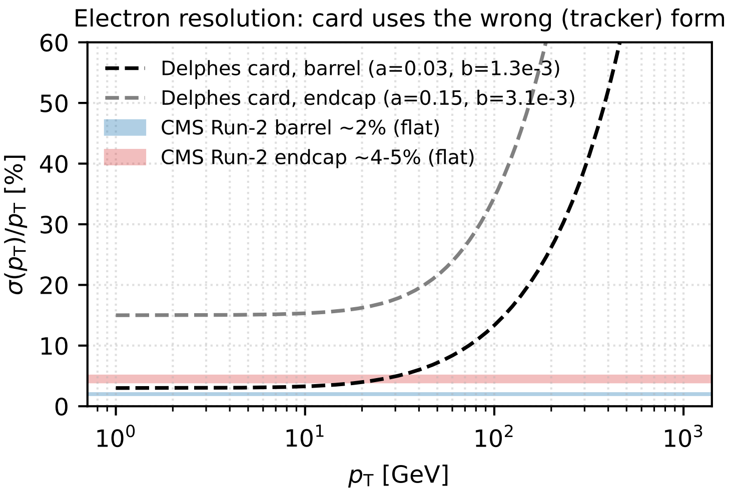

Figure 3: Electron fractional resolution. The card (dashed) applies the tracking form of Eq. 1, whose $b\,p_{\mathrm{T}}$ term makes the resolution diverge at high p_{T}; the measured CMS electron resolution [8] (bands) is approximately flat at ∼2% (barrel) and ∼4–5% (endcap). The $b\,p_{\mathrm{T}}$ term is a copy of the charged-hadron tracker coefficients and should be removed for electrons.

3. Several efficiency plateaus are optimistic. The ATLAS charged-hadron tracking efficiency plateau of 0.95 (central) and 0.85 (forward) sits above the measured [7] values of ∼0.86 and ∼0.73; the flat 0.95/0.85 identification efficiencies do not correspond to any single published working point, and depending on the intended WP are optimistic (loose-ID plateaus) or pessimistic (Medium/Tight muon WPs reach ∼0.98). These are absolute shifts of 5–10% that directly bias acceptance.

4. Some blocks are already correct — and should be locked in. The ATLAS electromagnetic-calorimeter resolution ($S = 10.1\%\sqrt{\mathrm{GeV}}$, $C = 0.17\%$ local; FCal $S = 28.5\%\sqrt{\mathrm{GeV}}$, $C = 3.5\%$) is confirmed against the test-beam and fast-simulation references [11, 12], and the b-tagging 70% working point with its c- and light-mistag rates is confirmed against the Run-2 performance measurement [13]. Recording these as *verified* is as valuable as flagging the broken ones: it tells a user which numbers they can trust.

4 Proposed revisions

Table 2 lists the headline source-faithful values for the blocks where a drop-in revision is well defined. The complete set of 88 proposed coefficient changes, each with source, locator, uncertainty, and verifier verdict, is released in the machine-readable record (`run1_provenance.json`) and the human-readable `findings_summary.md`. Consistent with the project's no-silent-overwrite rule, the original stock cards are left untouched; the proposed values are instead emitted into separate `*_baseline` (central values) and `*_uncertainty` (per-bin error bands) cards. This first-version (v1) pair applies the momentum-resolution revisions and the ATLAS charged-hadron tracking blocks; the efficiency, calorimeter, and tagging revisions are documented but not yet applied, pending the block-by-block review that the validation overlays (Appendix A) support.

Table 2: Representative proposed revisions (source-faithful central values). Full list with uncertainties in `run1_provenance.json`.

Quantity	**Region**	**Card**	**Proposed**
CMS charged-hadron mom. res. a / b (10^{-4} GeV^{-1})			
a, b	$\lvert\eta\rvert \leq 0.5$	0.06, 13	0.009, 2.3
a, b	0.5–1.5	0.10, 17	0.015, 4.2
a, b	1.5–2.5	0.25, 31	0.023, 9.5
Electron mom. res. (constant term; remove $b\,p_{\mathrm{T}}$; Run-2 sources)			
a (ATLAS)	$\lvert\eta\rvert \leq 0.5$	0.03	0.010
a (CMS)	$\lvert\eta\rvert \leq 0.5$	0.03	0.020
Efficiency plateaus			
ATLAS ch.-had. track	central, high p_{T}	0.95	0.86
ATLAS ch.-had. track	forward, high p_{T}	0.85	0.73
ATLAS muon ID	$\lvert\eta\rvert \leq 1.5$	0.95	0.98 (Medium WP)
ATLAS τ (1-prong)	–	0.70	0.55 (Medium WP)
Calorimeter			
CMS HCal S	$\lvert\eta\rvert \leq 3.0$	1.50	1.10

5 Validation and limitations

Three limitations are inherent to a literature-extraction approach and are recorded with every value. (i) Many resolution curves are published only as figures, so the corresponding numbers are *plot-reads* with a reading uncertainty of order one minor grid division; these are tagged as such. (ii) The Delphes η binning (boundaries at $0.5, 1.5, 2.5$) frequently matches *neither* detector's geometry (ATLAS: $1.05, 1.7, 2.0$; CMS: $0.9, 1.2, 2.4$), so some coefficients are necessarily interpolations across a boundary rather than a verbatim table value. (iii) Working-point ambiguity: a flat efficiency in the card maps to different published numbers depending on the intended identification WP, which the card does not state; we record the WP we assumed. (Three CMS blocks — b-tagging, electron momentum resolution, and τ-tagging — lost their verification stage to a workflow interruption and were re-verified in a separate pass; all three returned accept, so the record is now complete for all 26 blocks.)

6 Conclusion

A complete provenance pass over the ATLAS and CMS Run-2 Delphes cards is feasible as a verified, multi-agent literature audit. The pass confirms a number of card values against their sources and, more importantly, identifies concrete defects — a track-momentum resolution too pessimistic by a factor $\sim$3–12 depending on p_{T}, an electron resolution modelled with the wrong functional form, and several optimistic efficiency plateaus — each tied to a citable measurement and an uncertainty. The deliverables are a machine-readable provenance file, a human-readable findings summary, first-version `*_baseline` / `*_uncertainty` cards, and this manuscript with a self-validating overlay appendix. Completing the remaining efficiency, calorimeter, and tagging revisions would let Delphes studies carry a source-traceable detector-modelling systematic derived directly from the per-bin uncertainty bands.

A Validation of figure-read parameterizations

Several parameterizations are read from published performance figures rather than from a numeric table. To document that each extraction is faithful, we overlay the adopted Delphes curve (with its uncertainty band) directly on the source figure — so the reader can verify the agreement by eye, and so a mis-extraction is caught immediately. Where the source figure is published in an arXiv paper we draw on the rendered figure itself (Figs. 4–6); for ATLAS auxiliary figures we overlay on the digitized points, citing the figure (Fig. 7). Each panel reproduces the published figure under our curve; the band is the per-bin systematic used in the `*_uncertainty` card.

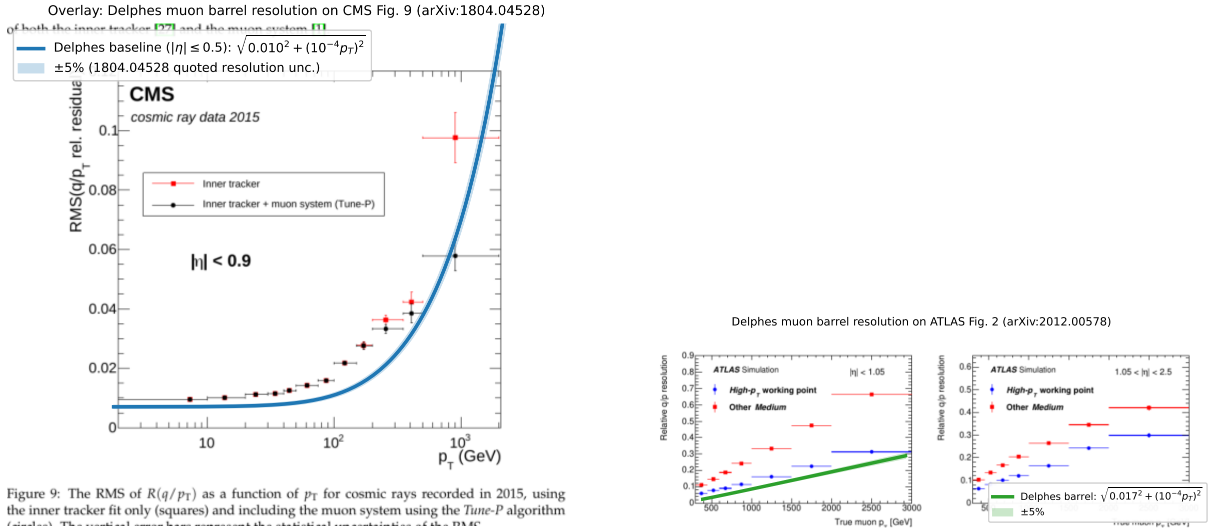

Figure 4: Muon momentum resolution. Left: Delphes CMS barrel curve on CMS Fig. 9 [5] (RMS of $R(q/p_\mathrm{T})$, $|\eta| < 0.9$, cosmic rays). Right: Delphes ATLAS barrel curve on ATLAS Fig. 2 [4] (q/p resolution vs. p_T, $|\eta| < 1.05$).

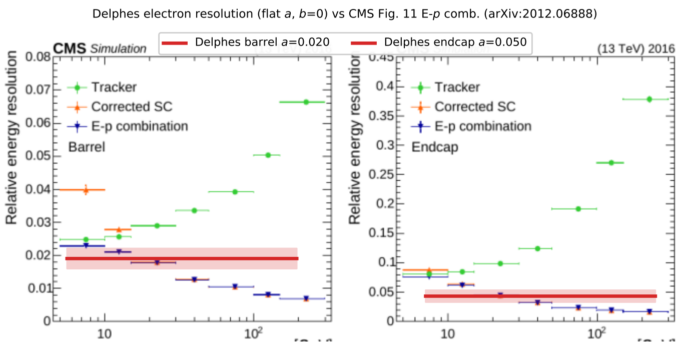

Figure 5: Electron resolution: the flat Delphes constant term ($b = 0$) on CMS Fig. 11 [8] (barrel/endcap relative energy resolution vs. p_T). Our value matches the E-p combination at low/intermediate p_T and is conservative at high p_T, where the calorimeter resolution improves.

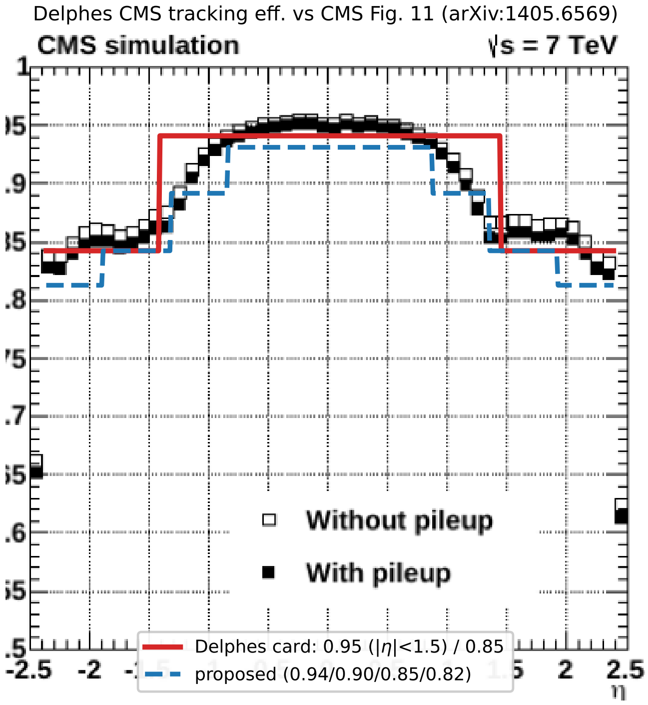

Figure 6: CMS charged-hadron tracking efficiency vs. η on CMS Fig. 11 [6]: the card step (0.95/0.85) and the proposed finer binning, both bracketing the measured plateau and forward fall-off.

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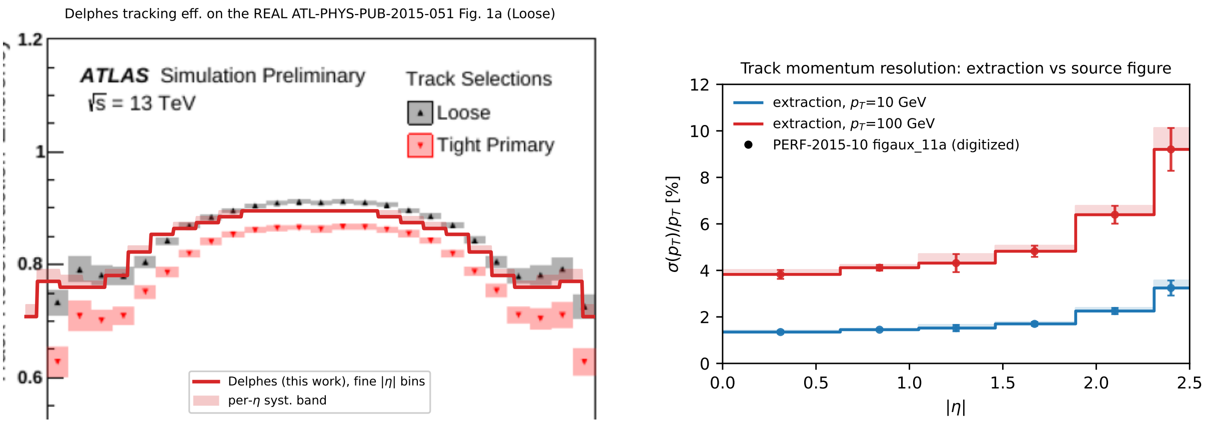

Figure 7: ATLAS charged-hadron tracking blocks. Left: DELPHES tracking efficiency (fine $|\eta|$ bins) on the real ATL-PHYS-PUB-2015-051 Fig. 1a (Loose selection) — the curve tracks the published points across η. Right: track momentum resolution vs. η (overlay on digitized points; the source auxiliary figure was not directly retrievable). Remaining blocks (muon/photon identification) are validated by the consistency arguments in the text (the muon plateaus of $\sim$0.99/0.95 make the card's flat 0.95 conservative).

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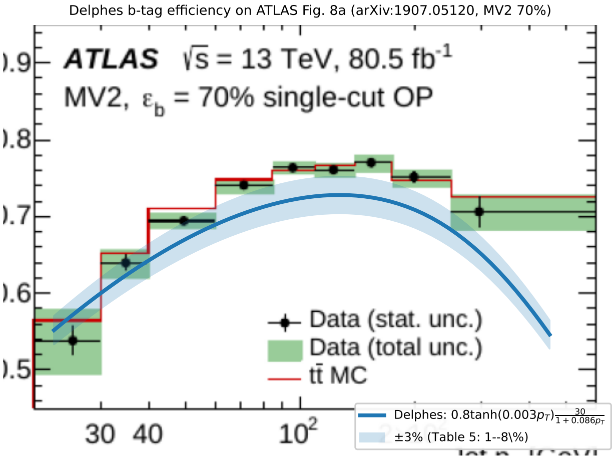


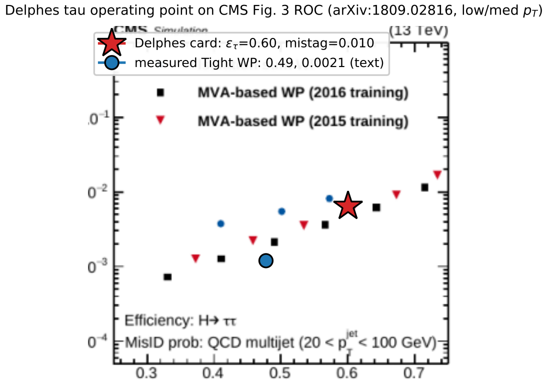
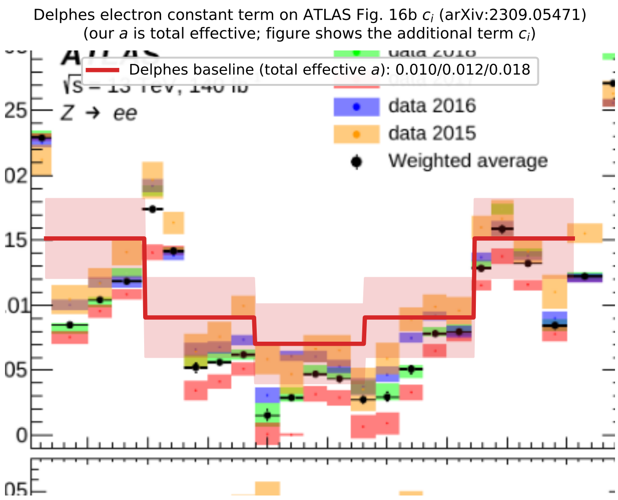


Figure 8: Flavour tagging. Left: DELPHES b-tagging efficiency on ATLAS Fig. 8a [13] (MV2 $\varepsilon_b = 70\%$); the card matches at low p_T but falls off too steeply at high p_T (a tuning target). Right: the card τ operating point ($\varepsilon_\tau = 0.6$, mistag $= 0.01$) on the CMS misidentification-vs-efficiency ROC, Fig. 3 [15]; the card point lies *above* the measured MVA curve — the flat 0.01 mistag is a factor $\sim$2–3 larger than the measured value at $\varepsilon_\tau = 0.6$, i.e. the card is conservative (pessimistic) on the τ mistag rate.



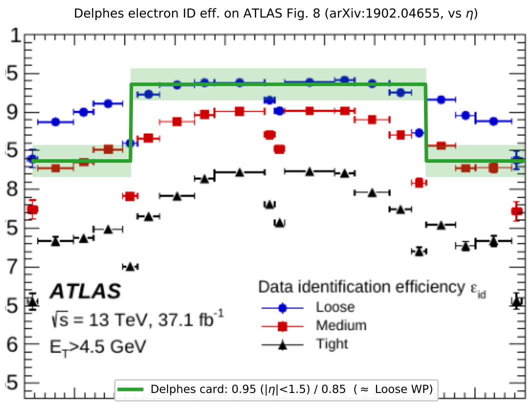


Figure 9: ATLAS electron. Left: the DELPHES electron constant term on the additional-constant-term $c_i(\eta)$ of Fig. 16b [10] (our a is the total effective term, so it brackets c_i from above and follows the same η structure). Right: DELPHES electron identification efficiency on ATLAS Fig. 8 [9] (vs. η); the card 0.95/0.85 is closest to the Loose working point, but does not match a single published WP — it is slightly optimistic centrally (Loose plateau $\sim$0.93) and pessimistic in the endcap ($\sim$0.90).

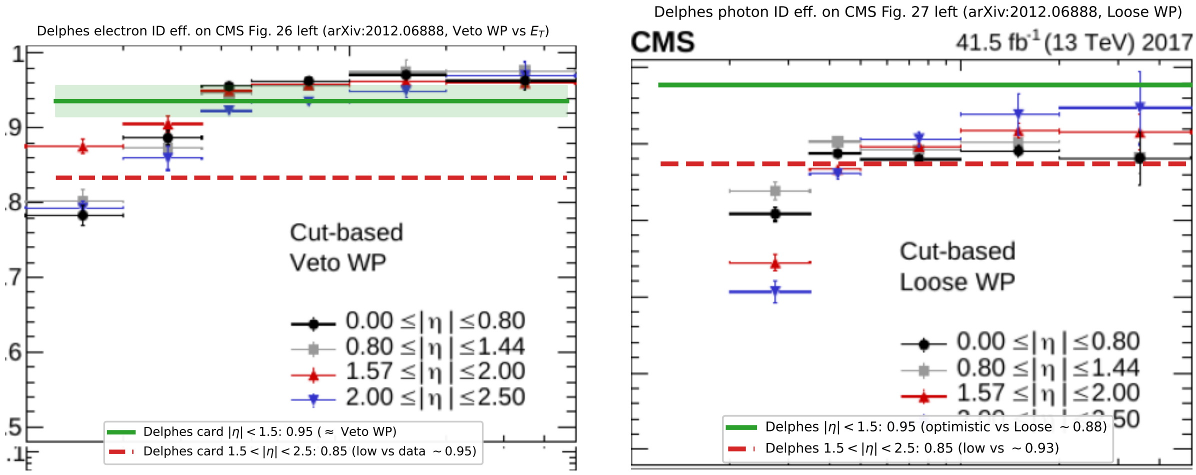


Figure 10: CMS electron (left) and photon (right) identification efficiency on CMS Figs. 26 and 27 [8] (cut-based working points vs. E_{T}, four η ranges). For electrons the card central 0.95 matches the Veto plateau while the forward 0.85 is too pessimistic (measured ∼0.95). For photons the card central 0.95 is mildly optimistic (Loose plateau ∼0.88) and the forward 0.85 low (measured ∼0.93) — both flagged for tuning.